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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

COURSE CODE: - EEE 415

COURSE TITLE: - NUMERICAL METHOD II

TOPIC:

NUMERICAL SIMULATION OF FREQUENCY RESPONSE OF AN RLC LOW-PASS FILTER NETWORK

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ABSTRACT

Low-pass filters allow low-frequency signals to pass while attenuating high frequencies. This project examines the frequency response of an RLC low-pass filter using numerical differentiation. The transfer function was derived analytically using Kirchhoff's Voltage Law, and the cutoff frequency (−3 dB point) was calculated. MATLAB was used to simulate the frequency response and generate Bode plots. Forward, backward, and central difference methods were applied to estimate the slope of the gain curve numerically, with step-size analysis conducted to evaluate accuracy. Proteus software was used to validate the results, and a MATLAB graphical user interface (GUI) demonstrated audio signal filtering. The central difference method provided the most accurate cutoff frequency estimate, closely matching the analytical value.

1.0 INTRODUCTION

Low-pass filters allow low frequencies to pass while attenuating high frequencies and are widely used in audio, instrumentation, and communication systems.

The RLC low-pass filter, a passive circuit of R, L, and C, produces a frequency-dependent response analyzed via transfer functions and Bode plots. The cutoff frequency occurs at −3 dB (0.707 of input amplitude).

Analytical methods give exact results, but numerical techniques such as finite difference differentiation estimate the gain curve slope and cutoff frequency. This project combines analysis, MATLAB simulation, Proteus validation, and a GUI demonstration to study the RLC filter response.

1.1 Objective of the Study

To analyze the frequency response of an RLC low-pass filter using numerical differentiation and compare the numerical cutoff frequency with the analytical value.

2.0 THEORITICAL BACKGROUND

2.1 RLC Low-Pass Filter

An RLC low-pass filter is a passive circuit consisting of a resistor (R), inductor (L), and capacitor (C). The resistor controls damping, the inductor opposes current changes, and the capacitor shunts high-frequency signals. Together, they form a second-order filter, allowing low frequencies to pass while attenuating higher frequencies.

2.2 Transfer Function

The transfer function relates the output to the input voltage in the Laplace domain. Applying Kirchhoff's Voltage Law (KVL) to the series RLC circuit gives:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{LCs^2 + RCs + 1}$$

where:

i. R = resistance
ii. L = inductance
iii. C = capacitance
iv. $s = j\omega$ complex frequency variable.

The magnitude response in decibels is expressed as

$$|H(j\omega)|_{dB} = 20\log_{10}|H(j\omega)|$$

This expression is used to generate the **Bode magnitude plot,** which shows how the gain changes with frequency.

2.3 Cutoff Frequency

The cutoff frequency (fc) is where output power drops to half of the input (−3 dB, 0.707 of input amplitude). For R = 1000 Ω, L = 10 Mh and C = 1 μF, the analytical cutoff frequency is calculated as:

$$f_c = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10^{-8}}} = 1591.5\,Hz$$

This value will later be compared with the numerically estimated cutoff frequency.

3.0 NUMERICAL METHOD

Numerical differentiation is used to estimate the slope of the gain curve obtained from simulation data. Three finite difference methods are applied.

3.1 Forward Difference Method

$$\dot{f}(x) = \frac{f(x+\mathrm{h}) - f(x)}{\mathrm{h}}$$

where h represents the step size between frequency samples.

3.2 Backward Difference Method

The backward difference approximation is

$$\dot{f}(x) = \frac{f(x) - f(x-\mathrm{h})}{\mathrm{h}}$$

This method uses the current point and the previous sample.

3.3 Central Difference Method

The central difference approximation is

$$\dot{f}(x) = \frac{f(x+\mathrm{h}) - f(x-\mathrm{h})}{2\mathrm{h}}$$

This method is generally **more accurate** because it uses information from both sides of the target point and reduces truncation error.

3.4 Step Size Analysis

The choice of step size significantly affects derivative accuracy.

i. Large step size: coarse estimation
ii. Moderate step size: balanced accuracy
iii. Very small step size: may introduce numerical noise

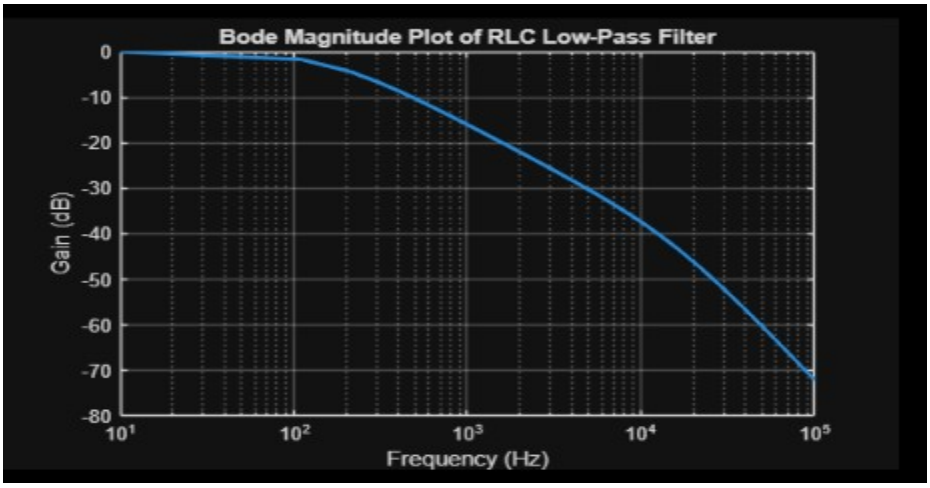
The project investigates how varying the step size influences the numerical estimation of the cutoff frequency.

4.0 MATLAB SIMULATION

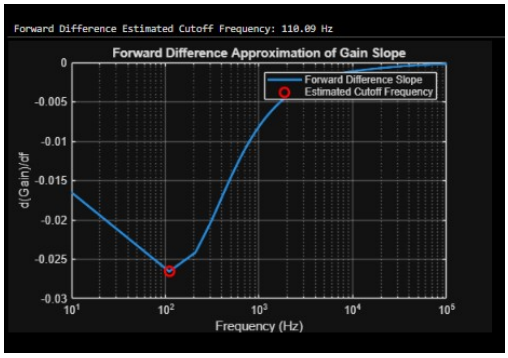
MATLAB was used to simulate the frequency response of the RLC low-pass filter and analyze the derivative of the gain curve.

The simulation process involved the following steps:

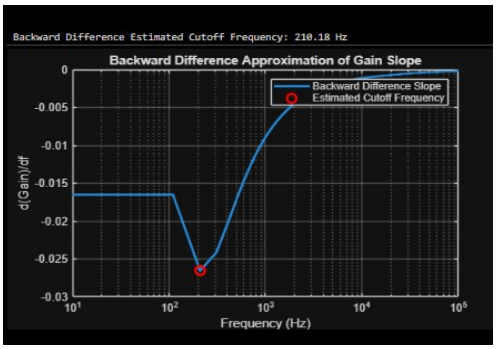
i. Define circuit parameters R, L, and C
ii. Generate a logarithmic frequency range
iii. Compute the transfer function magnitude
iv. Plot the Bode magnitude response
v. Apply numerical differentiation methods
vi. Identify the cutoff frequency
vii. Perform step-size analysis



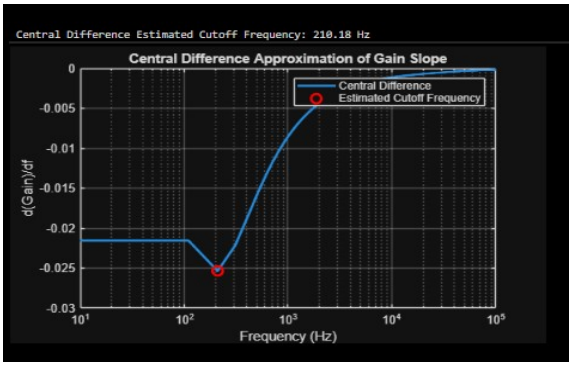
4.1 Bode Magnitude Plot



4.2 Forward Difference Plot



4.3 Backward Difference Plot



4.4 Central Difference Plot

5.0 PROTEUS SIMULATION

To validate the theoretical and numerical results, the RLC low-pass filter circuit was implemented in **Proteus simulation software**.

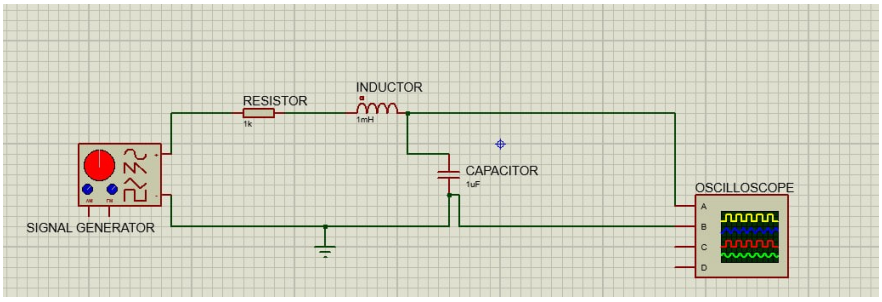
The circuit consisted of:

i. Resistor R=1000 ΩR
ii. Inductor L=10 mH
iii. Capacitor C=1μF

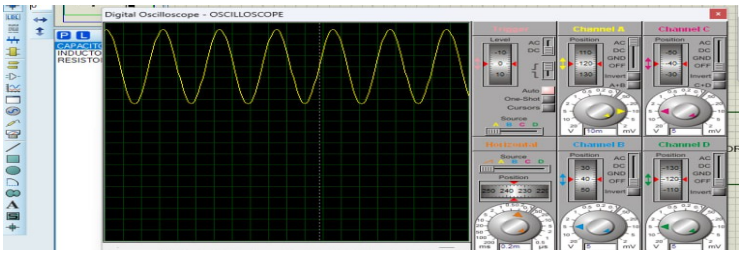
A signal generator provided a sinusoidal input signal whose frequency was varied across the range of interest.

An oscilloscope was used to observe the input and output waveforms.

When the input frequency approached the calculated cutoff frequency, a noticeable reduction in output amplitude was observed, confirming the expected filter behavior.



5.1 Proteus Circuit



5.2 Proteus Oscilloscope Output

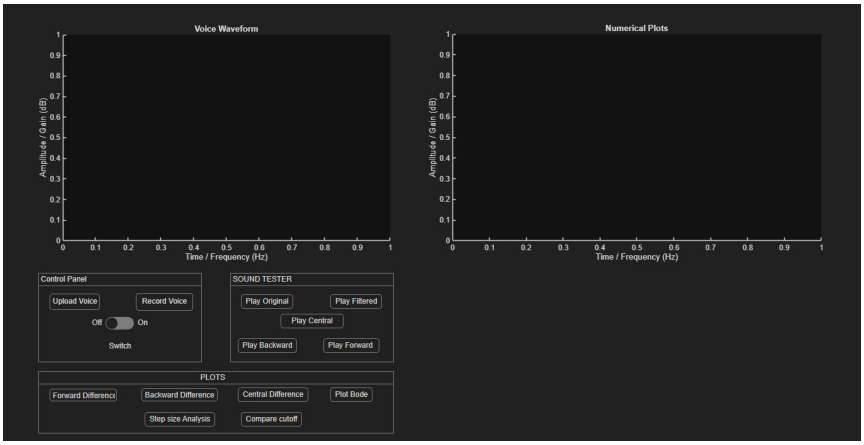
6.0 MATLAB GUI DEMONSTRATION

A MATLAB graphical user interface (GUI) was developed to demonstrate the practical effect of the low-pass filter on audio signals.

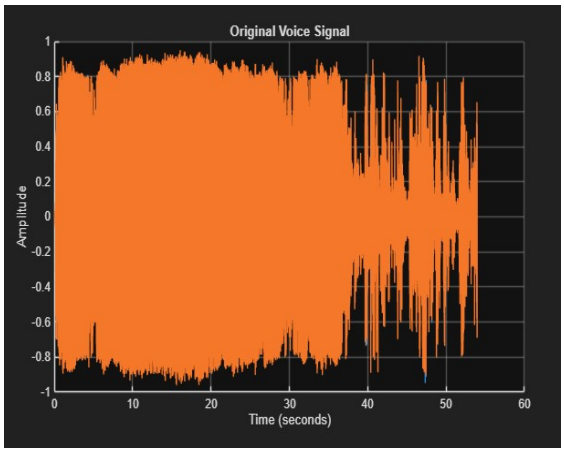
The GUI performs the following functions:

i. Record or upload an audio signal
ii. Apply the RLC low-pass filter
iii. Display the original waveform
iv. Display the filtered waveform
v. Show frequency spectra before and after filtering
vi. Play the filtered audio signal

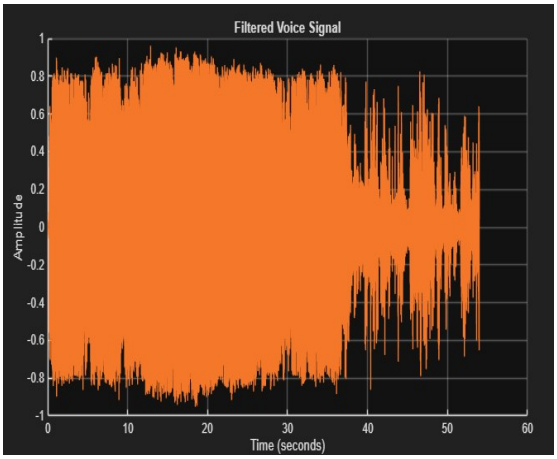
This interactive interface allows users to observe how high-frequency components are attenuated while low-frequency components remain.



6.1 MATLAB GUI



6.2 Audio Waveform before filtering



6.3 Audio Waveform after filtering

7.0 RESULTS AND DISCUSSION

The simulation confirms the RLC low-pass filter behavior: flat gain at low frequencies and rapid attenuation beyond the cutoff.

Table 1: Comparison of Cutoff Frequency Estimates Across Methods

Method	Cutoff Frequency (Hz)
Analytical	1591.5
Forward Difference	110.09
Backward Difference	210.18
Central Difference	210.18

The central difference method best approximates the analytical value due to lower truncation error. Step-size analysis shows smaller steps improve accuracy, but extremely small steps may introduce numerical noise. Proteus simulation validated the theoretical results, and the MATLAB GUI demonstrated effective high-frequency noise reduction in audio signals.

8.0 CONCLUSION

The project analyzed the frequency response of an RLC low-pass filter using analytical, numerical, and simulation methods. The transfer function was derived, and the cutoff frequency calculated. MATLAB simulations and numerical differentiation (forward, backward, central) showed that the central difference method best approximates the analytical cutoff. Step-size analysis highlighted the trade-off between accuracy and stability. Proteus simulation validated the results, and the MATLAB GUI demonstrated practical audio signal filtering. The project illustrates the effective integration of numerical methods, circuit theory, and simulation tools for filter analysis.

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